

Task 5: Exploratory Data Analysis (EDA)

Objective

EDA helps understand datasets by summarizing main characteristics and visualizing variable relationships.

It includes summary statistics, correlation analysis, and identifying trends, patterns, and anomalies.

Implementation

- Load dataset and compute **summary statistics**.
 - Check for **missing values** and handle them.
 - Create **correlation matrix** and visualize with a heatmap.
 - Plot histograms and boxplots for distribution analysis.
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Python Code

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
df = pd.read_csv("data.csv")

# Summary statistics
print(df.describe())

# Correlation heatmap
sns.heatmap(df.corr(), annot=True, cmap='Blues')
plt.title("Correlation Heatmap")
plt.show()

# Distribution plot
sns.histplot(df['Age'], kde=True)
plt.title("Age Distribution")
plt.show()
```

Client Project

Perform EDA on client-provided data to generate insights using visual and statistical analysis.